Social and Cultural Issues in Requirements Engineering

Handout # 7

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Introduction - 1

- Some aspects of the requirements engineering process deal with social and cultural issues
- What is the best way to deal with these issues?

Social Issues

Social Issues in RE - 1

- Requirements engineering is a social process, as it involves interaction among clients, engineers, and other systems

Social Issues in RE - 2

- Requirements engineering is not an entirely formal process, because it involves discovering client needs and reconciling them with technical possibilities

Stakeholders in RE Process

- At least three major groups participate in requirements engineering process;
  - The client organization
  - The requirements team
  - The development team
- There may be other interested parties, e.g., regulatory authorities

Areas of Social Issues

- Within the client organization
- Within the requirements team
- Between the client and the requirements team
- Between the development and requirements teams

Issues within the Client Organization - 1

- In a large organization, there are usually competing divisions or groups, so the notion of *'the client'* is not obvious
- Intended users of the system may be different people from the ones who interact with the requirements team (Hospital managers vs Laboratory Operators)

Issues within the Client Organization - 2

- The users of the system should be brought into the requirement engineering process, as they hold the key to the eventual success of the software engineering project
- They will be the people who use the system and eventually decide on its success or otherwise.

Issues within the Client Organization - 3

- The requirement process reveals the problems within the client organization, which must be addressed by facilitating communication among different stakeholders (e.g. Sales vs Security,,---Production vs delivery)
- The problems within the client organization must not be buried, as they effect the implementation of the project

Issues within the Client Organization - 4

- The new automated system may have profound impact on how the business is conducted (e.g. switching from counter sales to online system---Online classes) or
- How information is classified within the organization
  - e.g. Who should maintain the student personal file? Admissions? Or Program office in IMSciences?

Issues within the Client Organization - 5

- Success of the project requires that every group within the organization understand different aspects of the new system (e.g. security or efficiency)

- Problems of tacit knowledge
  - Say-do problem

Issues within the Requirements Team

- How work is organized?
- What methods and notations are used?
- What team members think about organization and how jelled requirement team is?

Issues between Client Organization and Requirements Team

- Financial arrangements (e.g. delays, changes?)
- Ethical obligations (e.g. confidentiality)
- Legal safeguards (which laws? Client involved in illegal activity)
- Denial of information
- Management of changes

Issues between Development and Requirement Teams

- Development team needs to work very closely with the requirements team to resolve inconsistencies and to get details
- In some cases, requirements team may be disbanded or assigned other tasks

Cultural Issues in RE

Cultural Issues in RE

- Advances in the internet and communication technologies has enabled customers and developers to collaborate with each other in geographically and temporally dispersed environments

Cultural Issues in RE

There may be

- Time zones differences
- Language and terminology differences
- Religious and racial differences
- Ethical issues
- Political differences
- Differences in business environment

Example: A Billion

- Scientific community and US consider the following number to be a billion

1,00,00,00,000

- For the rest of the world, a billion is

10,00,00,00,00,000

Differences in Time Zones - 1

- Working hours of clients and developers may differ by eight hours or more
- Arranging phone calls and video conferences become a hassle as one party has to come to office very early or stay very late
- Analysts start assuming requirements

Language and Terminology Differences - 1

- Clients and developers may speak different languages or different dialects
- Requirements errors are introduced by not understanding other partner's language and terminology properly
  - E.g Some of us don't know what 20:00 hour means, and some would know its 8:00 o'clock at night.
  - Or "Its kicking it"

Language and Terminology Differences - 2

- People and government in the US, and worldwide scientific community consider the following number to be a billion

  1,00,00,00,000

- For the rest of the world, a billion is

  10,00,00,00,00,000

Language and Terminology Differences - 3

- Globally, people communicate with fellow citizens using sports lingo to convey certain situations and concepts, even in the business environment
  - "Fielding set hai"
- This can cause misunderstandings

Language and Terminology Differences - 4

- Use of the word ‘hockey’ in Pakistan and US means two different sports: ‘field hockey’ and ‘ice hockey’ respectively

Religious and Racial Differences

- Insensitive comments on religious and racial backgrounds of people involved in software engineering projects can become a major hindrance in the successful execution of the requirements engineering process
  - E.g. You cant use the word “Majjar” or “Sunni/Shia” for a certain stakeholder in a disgraceful manner

Ethical Issues

- Access to confidential client information
- Possibility of elimination of jobs
- Differences of opinions with the client on the project (e.g. Tax stealing)

Political Differences

- Differences in political ideologies and personal convictions can also lead to unprofessional environment in the execution of the requirements engineering process
  - E.g. Benazir income support program
- Some people do not want to work on military software programs

Differences in Business Environments

- Every society has its own culture within the business community, which must be understood for successful execution of the requirements engineering process
- *E.g. Jumma Namaz, Ramadan, Rings in Auctions*

Addressing Social and Cultural Issues

Addressing Social and Cultural Issues - 1

- Understand social and cultural issues and differences
- Avoid judgmental comments and offensive remarks on un-related views and beliefs of others
  - E.g. don't indulge in PTI vs PML discussion or religious one

Addressing Social and Cultural Issues - 2

- Create an environment of respect and professionalism
- Focus on discovering the needs of the customers
- Use state-of-the-art technology to facilitate activities in the requirements engineering process

Summary

- Requirements engineering is not a strictly formal process, but one which has social and cultural side effects
- Requirements engineers must understand different aspects of these issues and address them in the requirements engineering process

References

- Social issues in requirements engineering
  - Joseph A. Goguen